SECTION - V

TECHNICAL SPECIFICATION

3.1 SCOPE:

This Specification covers design, manufacture, assembly, shop testing, forwarding and delivery F.O.R. destination including unloading and handling at stores/site of 250KVA 33KV/415V and 100kVA 33KV/415V Station Transformers . The prices quoted shall be FIRM.

3.2 STANDARDS:

Except otherwise specified or implied herein the station transformers shall comply in all respects with the latest applicable IS-1180 part-I and CBIP manual on transformers.

3.3 TYPE AND RATING:

The transformers shall be of core type, three phase, oil immersed and shall be suitable for outdoor service under the service conditions indicated in clause 23.1 of this specification. The transformers are normally used as step down transformers.

The required rating electrical characteristics of station Transformers are furnished below:

a)	Type	:	Core type 3 phase oil immersed outdoor type.
b)	Rated KVA	:	250KVA 100KVA
c)	Service	:	Outdoor as step down
d)	No. of phases in the system	:	Three
e)	Frequency	:	50 Hz.
f)	i) Double wound or Auto Transformers	:	Double wound
	ii)Winding conductor material	:	Copper
	iii) Winding connections	:	a) HV Delta b) LV Star
g)	Type of cooling medium	:	Mineral oil

h) Rated Voltage:
   i) Higher voltage winding for 33KV : 36.3 kV
   ii) Lower voltage winding : 415 Volts

i) Highest System Voltage :
   i) Highest System Voltage for 33KV : 36.3kV

	ii)	Lower Voltage System	:	As per IS:1180
j)	Method of System Earthing		:	Solidly Earthed Neutral
k)	Insulation level of winding		:	
	a)	33 kV windings	:	170 kV
	b)	Lower Voltage winding	:	As per IS:1180
l)	Power Frequency voltage			
	a)	33 kV Windings	:	70 kV
	b)	Lower Voltage winding	:	As per IS:1180
m)	Vector relation ship required between windings and Vector symbol		:	Dyn11 Dyn11
n)	Tap changing		:	off circuit with pad lock arrangement

o)	Tapping shall be provided in Higher voltage winding	:	250KVA 100KVA
	Tap Range	:	± 10% ± 10%
	No.of Taps	:	9 9
	Each Tap	:	± 2.5% ± 2.5%
	Normal Tap	:	5 5

p)	Type of Cooling	:	ONAN
q)	Impedance Voltage at 75 ^{deg} C with (Tolerance as per/ IS: 1180 part-I)	:	4.5% 4.5%
r)	Stabilizing Winding	:	Not required
s)	Whether Neutral Terminals are to be brought out:		
	i) Higher Voltage winding	:	No
	ii) Lower Voltage winding	:	Yes

t)No load current shall be as per IS 1180 latest

u):Limits of temperature rise shall be as per IS 1180 latest.

v):Bushing creep age shall be mentioned as per requirement.

w): Core shall be latest IS 3024 and major materials shall be sec 9.1 as per IS 1180

x): The tolerances on Electrical Performances excluding losses shall be as per IS 1180 (Part-I).

3.4 GUARANTEED LOSSES:

3.4.1 Maximum Losses

While the Bidders may offer their design, with guaranteed transformer losses at 75 deg. C. should not exceed the following maximum limits which include tolerances as per IS :

Energy Efficiency Level-1			
S.No.	Rating of Transformer	50Percent of Load	100Percent Load
1	100kVA,33KV/415V	510.63w	1773.75w
2	250kVA,33KV/415V	1053.5w	3149.75w

LT Transformer losses shall not exceed the value specified above failing which, the LT Transformer shall be rejected.

3.4.2 Bidders may please note that the maximum ceiling limit for transformer losses for station transformers of various voltage class and KVA rating shall be as indicated as above which include necessary tolerances as may be applicable as per relevant Bureau of Indian Standard (BIS) or any other equivalent International Standard. (No further tolerance in the ceiling limit for all losses therefore shall be permitted).

3.4.3 **Bids offering for transformers with losses in excess of ceiling limits mentioned under clause 3.4.1 shall be treated as non-responsive and rejected.**

Capitalization of losses will not be allowed even if the bidder quotes for less than the specified losses in the bid.

3.4.4 The Bidder must clearly specify that the offered losses are "FIRM" and no tolerances as per IS shall be applicable on the offered values.

3.4.5 As described above, the bidders are permitted to quote the same or lower to the maximum losses specified which will be considered on equal basis.

3.4.6 GUARANTEED LOSSES AND PENALTY :

The actual no load loss in kilowatts at rated voltage and rated frequency and the total load losses in kilowatts at rated output, rated voltage and rated frequency at 75°C shall be within the guaranteed losses. It may be noted that the actual losses shall not exceed the guaranteed losses by the bidder during final testing of the transformer under any circumstances.

During the testing if the actual losses are higher than the specified loss levels, the material will be rejected .

3.4.7 The temperature rise of the windings, core and oil of the transformer shall not exceed the specified limits in latest IS:1180 and the guaranteed values shall be furnished in the Tender submitted.

3.5 CORE:

3.5.1 The core shall be built up with high grade non-ageing, low loss and high permeability CRGO Silicon, steel laminations specially suitable for transformer cores. The laminations shall be treated to remove all burrs and shall be reannealed to remove all residual stresses. Each lamination shall be coated with a durable, black enamel insulation coating which shall be inert to the action of hot transformer oil/Carlite coating on CRGO allowed. Paper and varnish insulation will not be accepted. The nature of insulation should specified in the bid.

The core shall be tightly clamped and bolted to ensure adequate mechanical strength and to prevent vibration during operation. The bolts used in the assembly of the core shall be suitably insulated and clamping structure shall be so constructed that eddy current will be minimum.

The core shall be provided with lifting lugs suitable for lifting the complete core and coil assembly of the Transformer.

The core and the coil shall be so fixed in the tank that shifting will not occur when the transformer is moved or during a short circuit.

3.5.2 The transformers shall be suitable for over fluxing (due to combined effect of the voltage and frequency) upto 12.5% on any tapings without core heating. The maximum flux density in any part of the core under such conditions shall not exceed 1.9 tesla.

3.6 WINDING:

3.6.1 The windings shall be so designed that all the coil assemblies of identical voltage ratings shall be interchangeable and field repairs to the windings can be made readily without special equipment. The coils shall be supported between adjacent sections by insulating spacers and barriers, bracings and other insulation used in the assembly of the windings shall be treated with suitable insulating varnish or equivalent compound to develop the full electrical strength of the windings. All materials used in the insulation and assembly of the windings shall be insoluble non-catalytic and chemically inactive in the hot transformer oil and shall not soften other-wise be adversely affected under the operating conditions.

3.6.2 All threaded connections shall be provided with locking facilities. All leads from the winding to the terminal board and bushings shall be rigidly supported to prevent injury from vibration.

3.6.3 The windings shall be clamped securely in place so that they will not be displaced or disturbed during short circuits. The assembled core and windings shall be vacuum dried and suitably impregnated before removal from the treating tank. The conductors used in the coil structure shall be best suited to the requirements and all permanent current carrying joints in the windings and the loads shall be welded or brazed.

3.7 INSULATING OIL:

The oil for first filling shall be supplied with each transformer. The oil shall comply in all respects with the provisions of latest IS:335.

3.8 TANK:(Construction shall be as per latest IS 1180)

3.8.1 The transformer tank and cover shall be fabricated from good commercial grade low carbon steel suitable for welding and of adequate thickness. The tank shall be provided with bolted top cover. All tank gaskets used shall be of NBR (Acrylonitrile Butadiene Rubber) suitable for temperature conditions expected to be encountered during operation. The main tank body shall have sufficient strength to withstand without permanent distortion (1) filling by vacuum at 750 mm mercury without oil and (2) continuous internal gas pressure of 0.35 atmospheres up to operating level of oil. The tank cover shall be bolted to the tank and the transformer design shall be such that the tank will not be split between the lower and upper cooler connection for untanking.

3.8.2 All bolted connections to the tank shall be fitted with suitable oil tight gaskets which shall give satisfactory services under the operating conditions. Special attention shall be given to the method of making the hot oil tight joints between the tank and the cover, also between the cover and the bushings and all other outlets to ensure that the joints can be remade satisfactorily with the help of

semiskilled labor. Where compressible gaskets are used, steps shall be provided to prevent over compression. Gaskets used shall confirm to Type-III as per IS.1149/Type-C as per IS-4253 Part-2

3.8.3 Suitable guides shall be provided for positioning the various parts during assembly or dismantling. Adequate space shall be provided between the cores and the windings and the bottom of the tank for collection of any sediment.

3.8.4 Lifting eyes or lugs shall be provided on all parts of the transformer requiring independent handling during assembly or dismounting. In addition, the transformer tank shall be provided with lifting lugs properly secured to the sides of the tank, for lifting the transformer either by crane or by jacks.

3.8.5 The design of the tank, the lifting lugs and boxes shall be such that the complete transformer assembly filled with oil can be lifted with the use of four lugs without any damage or distortions.

The tank shall be provided with two suitable copper alloy lugs for the purpose of grounding.

3.8.6 One filter valve on the upper side of tank(above 200kVA shall be as per section 20.1 of IS 1180.

3.8.7 The transformer tank shall be supported on a structural steel frame with forged steel or cast steel single flanged wheels suitable for moving the transformer completely filled with oil as per latest IS 1180.

3.9 TAPS:

3.9.1 Tapings shall be provided in high voltage with tap range ± 10% winding to obtain a voltage variation insteps of ±2.5%. The transformer shall give full load on all taps.

3.9.2 Tap changing shall be carried out by means of an externally operated self positioning switch and when the transformer is in de-energized condition. Switch position No.1 shall correspond to the maximum plus tapings. Provision shall be made for locking the tapping switch handle in position.

3.10 OIL PRESERVING EQUIPMENT:

3.10.1 The conservator shall have a shut off valve and sump with a small drain valve and stop cock.

3.10.2 A Silicagel breather shall be provided for each conservator vessel. The external atmosphere shall not be in continuous contact with the Silicagel.

3.11 BUSHINGS;

3.11.1 The bushings shall have high factors of safety against leakage to ground and shall be so located as to provide adequate electrical clearances between bushings and grounded parts. Bushings of identical voltage and current rating shall be inter changeable. All bushings shall be equipped with suitable bushes of approved type and size and all external current carrying contact surfaces shall be shall be either Electro Tinning/silver plated adequately..

3.11.2 All the main winding leads shall be brought out through outdoor type bushings which shall be located such that the full flash over strength will be utilized and the adequate phase clearance shall be realized.

3.11.3 Each bushing shall be so co-ordinated with the transformer insulation, that all flash-overs will occur outside the tank.

3.11.4 All porcelain used in bushings shall be of the wet process, homogenous and free from cavities or other flaws. The glazing shall be uniform in colour and free from blisters, burrs and other defects. The interior surface shall be painted with hot oil resistant paint. The thickness of painting and paint scheme shall confirm to table-12 of IS 1180-1 Section 15.5 ,paint 631 of IS 5.

3.11.5 All bushings shall have puncture strength greater than the dry flash over value. The spacing between the bushing shall be adequate to prevent flash over between phases under all conditions of operation.

3.11.6 The bushings used shall comply with IS:2099 and the dimensions for the following voltage classes shall conform to latest ISS mentioned against them.

Voltage classes	Indian standard Specification *for porcelain parts.*		For metal parts	
36 kV	IS:3347	Part-V Section 1 1979	IS:3347	Part-V Section 2 1979
Up to 1.0 KV bushings	IS 3347 Part 1/Sec-I		IS 3347	Part I/Sec-2

3.12 CENTRE OF GRAVITY:

The centre of gravity of the assembled transformer shall be low and as near the vertical centre line as possible. The transformer must be stable with or without oil.

3.13 ACCESSORIES:

The supply shall include the following with each transformer.

a) First filling of oil conforming to IS:335.

a) FITTINGS AND ACCESSORIES:

i) Rating shall be as per BIS I of latest IS 1180 and terminal making Plates shall be as per BIS 4 of latest 1180.

ii) Earth terminals .. 2 Nos.

iii) Lifting lugs.

iv) Main oil guage.

vi) Drain cum sampling valve with plug or cover plate shall be as per section 20.1 latest IS 1180

vi) Thermometer pocket with cap.

vii) Air release plug.

viii) Oil conservator with drain plug and oil filling hole with blanking flange.

ix) Pressure relief Vent or Presuure relief valve.

x) Silicagel breather.

xi) Top filling valve with plug or cover plate (32mm dia.)

xii) Mounting channel and plain rollers for 100KVA shall be as per BI 86 for 250KVA shall be as per BI 87 of latest IS 1180

xiii) Lock with keys in duplicate for externally operated tap switch.

xiv) H.V.Outdoor bushings.

xv) L.V.Outdoor bushings.

xvi) Oil level indicator shall be 20.1b of latest IS1180

xvii) Terminal connectors shall be provided.

xviii) Inspection hole(For 200 KVA above shall be as per latest IS 1180)

xix) One filter valve on the upper side of tank(above 200kVA shall be as per section 20.1 of IS 1180

The transformer shall be supplied with all the accessories and fittings for normal operation whether they are mentioned above or not and the transformer shall be complete in itself.

3.14 PAINTING:

Before despatch all steel works not under oil shall be painted with a primary coat of anti-corrosive paint of durable nature and two coats of dark admiral grey paint. The interior surfaces shall be painted with hot oil resistant paint. Paint scheme shall confirm to table-12 of IS 1180-1 Section 15.5 ,paint 631 of IS 5 and according to the manufacturers standard practice.

3.15 PACKING:

The packing may be in accordance with bidder's standard practice to withstand the transportation.

3.16 ASSEMBLY AND TESTING:

3.16.1 The transformer shall be completely assembled and tested at the factory in accordance with IS:1180. The tests which shall be performed on the transformers by the manufacturers shall be indicated in the Tender.

3.16.2 Three copies of each test report shall be furnished for approval.

3.17 TERMINALS:

H.V. Transformer terminal shall be suitable for connecting Zebra/Deer conductor and L.V. terminals for connecting PVC Aluminium power cable(.HV side:Transformer terminal shall be suitable for connecting Zebra/Deer conductor
LV Cable Size:
100KVA-3.5C 95 Sqmm
250KVA-3.5C 120 sqmm)
(Any other details will be furnished at the time of detailed engineering)

3.18 CLEARANCE:
The minimum clearance in air shall be as follows as per IS-1180

Voltage	Medium	Clearance phase to phase in mm		Clearance phase to earth in mm	
		Open	Cable box	Open	Cable box
415 volts	AIR	75	25	40	20
33kV	AIR	350	350	320	220

3.19 DRAWINGS:

3.19.1 The manufacturer shall supply four copies of the drawings which will describe the equipment in detail for approval. After approval six complete sets of final drawings for each substation shall be sent. The reproducible drawings may also be supplied.

3.19.2 The following drawings for each item are to be supplied as a part of the contract.

a) Outline dimensional drawings of transformer and accessories.

b) Assembly drawings and weight of main components.

c) Drawings giving the weights for foundation.

d) Tap changing and name plate diagram.

e) Large scale drawings of high and low tension windings of the transformers showing the nature and arrangement of insulation and terminal connections.

f) Bushing drawing and Specification.

g) Test reports (for both type and routine tests).

3.19.3 Six copies of instruction booklets/operation and maintenance manuals shall be supplied for each substation. Descriptive literature and data on transformer, construction, winding, bushing, heat exchanger etc., shall be supplied for each transformer.

3.20 GUARANTEED TECHNICAL AND OTHER PARTICULARS:

The Guaranteed technical and other particulars of the transformers shall be furnished in the proforma given in annexure. Any tender which does not include the above particulars is liable to be rejected.

4.0 PARALLEL OPERATION:

The similar ratio transformers shall operate satisfactorily in parallel with each other, if connected between high voltage and low voltage bus bars.

5.0 TESTS

a) All the offered equipment shall be fully type tested by the bidder as per the relevant standard. Type test reports of similar/higher capacity of HV voltage class (Ratings as mentioned in IS:1180)along with an undertaking letter that bidder shall conduct & submit the complete type test reports for the offered design after award of contract, at bidder's cost without affecting the delivery schedule will be accepted for bid evaluation. The successful bidder have to submit the complete type test reports conducted at CPRI /ERDA and

Certification as per quality control GOI Order dated: 07.08.2015 for the offered design after award of contract at the suppliers cost without affecting the delivery schedule and before commencement of supplies. However it may be noted that ratings of station Transformers mentioned in IS:1180 will only be considered for Supply, Performance and Type tests for bid evaluation.

i) Impulse Voltage withstand Test
ii) Temperature Rise Test

b) Special tests other than type and routine tests, as agreed between Purchaser and Bidder shall also be carried out as per IS: 1180. with latest amendments.

c) The pre-shipment checks to be carried out by the Bidder are given under clause 6.0, "Inspection".

d) The bidder shall offer the transformer type tested. The type tests should have been conducted not earlier than 5 years as on date of bid opening.

e) The bidder shall conduct type tests at free of cost before commencement of supply in case the type tests are not carried out on the same design of the offered Transformer .

i) Type Tests: Validity of Type test is 5years as per latest CEA guidelines on Type test validity.

The Transformer shall confirm to all the type tests including following special test (as type test) in accordance with latest issues of IS: 1180:

a) Measurement of zero sequence impedance
b) Short circuit test
c) Measurement of acoustic noise level.This shall conform to NEMA standard publication TR-1
d) Measurement of capacitance and tan delta of transformer winding.

ii) Routine Tests (Acceptance tests):

1. Measurement of winding resistance
2. Measurement of Voltage ratio & check of phase displacement
3. Measurement of short circuit impedance (principal tapping when applicable) and load loss at 50% & 100% load.
4. Measurement of no load loss & current at 90%, 100% & 110% excitation as applicable.
5. Measurement of insulation resistance
6. Induced over voltage withstand test
7. Separate- source voltage withstand test
8. Repeat NLL after dielectric test
9. Pressure test as per 21.5 of IS:1180
10. Oil leakage test as per 21.5 of IS:1180
11. Tests on Transformer oil

    a) BDV
    b) Moisture content
    c) Resistivity
    d) Tan delta
    e) Interfacial tension.

AP Transco Engineers will witness all routine tests as mentioned above on all the Transformers.

iii) Tank Tests:

Following Tests (as Routine Test) shall be performed on tank (100%) as per CBIP manual on Transformers. AP Transco Engineer will witness one transformer tank per lot per KVA rating.

i) Vacuum tests
ii) Pressure test

iv) Winding:

The following tests shall be witnessed by AP Transco Inspector on one per lot per KVA rating.

1. Dimensional Verification
2. Measurement of DC resistance
3. One sample/winding will be selected for purity testing at in house/any NABL accredited lab.

v) Core building:

The following tests shall be witnessed by AP Transco Inspector on one per lot per KVA rating.

i) Check dimension (diameter, thickness, stack height and vertically)after core building – Testing.

ii) CRGO sheet selection during core building & assembly for testing of stacking factor , Magnetic polarization specific core loss, surface insulation resistivity, ductility & aging test during core stacking stage – Visual and Measurement.

iii) Traceability record of Parent coil - Verification.

vi) Oil Tests:

Test certificate shall be reviewed by the AP Transco inspector during FAT. Tests shall be conducted at NABL accredited lab as per IS:335 (latest amendment). End user name i.e. APTransco shall be mentioned on the TC.

6.1 Test reports

a) Four (4) sets of certified test reports and oscillograms shall be submitted for approval prior to despatch of the equipment. The equipment shall be dispatched only when all the required type and routine tests have been carried out and test reports have been approved by the Purchaser.

b) Four (4) copies of the test reports for the tests carried out on the ancillary apparatus shall be furnished to the Purchaser for approval prior to despatch.

c) All auxiliary equipment shall be tested as per the relevant standard. Test certificate shall be submitted for bought out items.

6.0 INSPECTION

6.1 General

i) The owner shall have access at all times to the works and all other places of manufacture where the transformers are being manufactured and the Bidder shall provide all

facilities for unrestricted inspection of the Bidder's works, raw materials, manufacture of all the accessories and for conducting necessary tests as detailed herein. The material offered for inspection should be ready by the offered date only

ii) The successful Bidder shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of equipment in its various stages so that arrangements could be made for inspection.

iii) No material shall be dispatched from its point of manufacture unless the material has been satisfactorily inspected and tested.

iv) The acceptance of any quantity of equipment shall in no way relieve the successful Bidder of his responsibility for meeting all the requirement of this specification and shall not prevent subsequent rejection if such equipment are later found to be defective.

6.2 Inspection programme

a) The Bidder shall chalk out a detailed inspection and testing programme for manufacturing activities for the various components. An indicative programme of inspection as envisaged by the Purchaser. This is not however intended to form a comprehensive programme as it is Bidder's responsibility to draw up and carry out such a programme duly approved by the Purchaser. The owner reserves the right to carryout any tests by a third party.

b) Cost of inspection/tests and third party inspection if any shall be borne by the Bidder.

c) Additional tests, if required, are deemed to be included in the scope of work.

d) Stages of inspection and Purchaser's participation would be defined and tied up at the time of award of contract.

7.0 QUALITY ASSURANCE PLAN :

7.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

a) The Structure of organization

b) The duties and responsibilities assigned to staff ensuring quality of work

c) The system of purchasing, taking delivery and verification of materials

d) The system for ensuring quality of workmanship

e) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 on ISO 9002 as applicable.

f) Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

g) Information and copies of test certificates as on (i) above in respect of bought out items.

h) List of manufacturing facilities available

i) Level of automation achieved and list of areas where manual processing exists.

j) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

k) List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vise-a-versa the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

7.2 The Supplier shall within 30 days of placement of order, submit the following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and Supplier before the QAP is finalized.

The Supplier shall submit the routine test certificates of bought out accessories and central excise invoice for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

7.3.1 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

a) Quality System:

* The Structure of the Organization
* The duties and responsibilities assigned to staff ensuring quality of work.
* The system of purchasing, taking delivery of verification of materials.
* The system of ensuring of quality workmanship.
* The system of control of documentation.
* The system of retention of records.
* The arrangement of Supplier internal auditing.
* A list of Administrator and work procedures required to achieve Suppliers quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

b) Quality Plans:

a. An outline of the proposed work and program sequence.
b. The structure of Suppliers organizations for the contract.
c. The duties and responsibilities ensuring quality of work

d. Hold and notification points.
e. Submission of engineering documents required by this specification.
f. The Inspection of the materials and components on request.
g. Reference to Suppliers work procedures appropriate to each activity.
h. Inspection during fabrication/construction.
i. Final inspection and test.

8.0 DOCUMENTATION:

8.1 i) The Bidder shall furnish along with the bid the dimensional drawings of transformer, GA and all other accessories.

ii) These drawings shall include the following information.

a) Detailed internal drawings of transformer clearly indicating insulation, core assembly and windings duly incorporating all important dimensions.

b) Tolerances on dimensions.

c) Material designation used for different components with reference to standards.

d) Fabrication details such as welds, finishes and coatings.

e) Catalogue or part numbers for each component and the total assembly with bill of materials.

f) Identification marking.

g) Weight of individual components and total assembled weight.

iii) The assembly drawings shall include the following:

a) Brief installation instructions.

b) Design installation torque for the bolt or cap screw.

c) Withstand torque that can be applied to the bolt or cap screw without failure of components or parts thereof.

d) Compression die number with recommended compression pressure.

e) Relevant technical details of significance.

8.2 Test Reports

i) Four sets of test reports shall be furnished to the Purchaser immediately after conducting the tests. one copy will be returned duly certified by Purchaser to the Bidder within three weeks thereafter and on receipt of the same Bidder shall commence with the commercial production of the concerned material.

ii) All records of routine test reports shall be maintained by the Bidder at his works for periodic inspection by the Purchaser.

iii) All test reports for tests conducted during manufacture shall be maintained by the Bidder. These shall be produced for verification as and when requested for by the Purchaser.

8.3 All drawings shall conform to International Standards Organization (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units only.

The Bidder shall within 2 weeks of placement of order, submit four sets of final versions of all the above drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Bidder within reasonable time. The Bidder shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval within two weeks from the date of Purchaser's comments.

After receipt of Purchaser's approval, the Bidder shall within three weeks, submit 6 prints and two good quality reproducible of the approved drawings for Purchaser's use.

8.4 Adequate copies of contract drawings, acceptance and routine test certificates, duly approved by the Purchaser shall accompany each of the dispatched consignment.

8.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Bidder's risk only.

8.6 Six copies of nicely printed and bound volumes of operation, maintenance and erection manuals in English language shall be supplied by the Bidder along with each Transformer to destination. The manual shall contain all the drawings and information required for erection, operation and maintenance of the transformers. The manual shall also contain a set of all the approved drawings, type, routine & acceptance Test reports etc.

8.6 Approval of drawings by the Purchaser shall not relieve the Bidder of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices. The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of ordering and the Purchaser shall have the power to reject any material which in his judgment is not in full accordance therewith.

9 PACKING AND FORWARDING:

9.1 The equipment shall be packed in crates suitable for vertical/horizontal transport as the case may be and suitable to withstand handling during transport and outdoor storage during transit. The Bidder shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing. The easily damageable material shall be carefully packed and marked with the appropriate caution symbol. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc. shall be provided. Any material found short inside the packing cases shall be supplied by Bidder without any extra cost.

9.2 Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee
b) Details of consignment
c) Destination
d) Total weight of consignment
e) Handling and unpacking instructions.

f) Bill of material indicating contents of each package.

The Bidder shall ensure that the packing and bill of material are approved by the Purchaser before despatch.

9.3 Weight of the transformer shall be designed, so that the transformer, can be transported by road. Heaviest single packet that can be transported by owner's trailer is about 60 MTs.

10.0 QUANTITY AND DELIVERY REQUIREMENTS :

i) This is set out in Schedule of Requirements Section-IV

ii) The scope of supply shall include a supply of 2.5% extra quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

ANNEXURE - 1

(GUARANTEED TECHNICAL & OTHER PARTICULARS)

SL. NO.	Particulars	
1.	Make	:
2.	Normal full load output (KVA)	:
3.	i)Type of Transformer :	
	ii) Winding type (HV & LV)	:
	iii) Material	:
4.	Type of cooling	:
5.	Indoor or Outdoor	:
6.	Rated high voltage (V)	:
7.	Rated low voltage (V)	:
8.	Connection and vector group	:
9.	Maximum temperature rise of	:
	i) Winding by resistance. (oC)	:
	ii) Oil by thermometer . (oC)	:
	iii) Maximum ambient air temperature .(oC)	:
10.	Energy Efficiency level	:
11.	a) Max total losses at 50% of rated load: b) Max total losses at 100% of rated load:	
12.	Impedance (Percentage)	:
13.	Tap changer	
	i) Make	:
	ii) Type	:
	iii) Indicator arrangement	:
	iv) Packing arrangement	:
	v) Other details	:
14.	Tapping on HV side for LV variation.	:

Tap Range :
No.of Taps :
Each Tap :
Normal Tap :

15. Impulse strength of HV winding :

16. Impulse strength of LV winding :

17. Overall dimensions of transformer in mm :

18. Approximate weight of :

    i) Core cum windings (Kg) :
    ii) Oil (Kg) :
    iii) Complete transformer with all fittings and including oil (Kg) :

19. Type of tank :

20. Make and type of breather fitted to the transformer :

21. Particulars of bushings :
    i) Country of origin :
    ii) Rating :
    iii) Types :
    HV bushings :
    LV bushings :
    iv) Impulse and flashover voltages :

22. Efficiency at u.p.f. and 0.8.p.f. :

23. Regulation at full load u.p.f. and 0.8.p.f. :

24. Terminal arrangements of bushings (HV and LV) :

25. Mounting arrangement for the transformer :

26. List of fittings and accessories provided. :